

EDUCATIONAL ANALYTICS • INTERNAL USE

Student Result Analysis Report

An evaluation of academic performance
across demographic and family-background
factors for 11,040 students.

COHORT SIZE	DATA SOURCE	PREPARED
11,040 Students	Student Result Analysis Notebook	29 August 2026

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Headline Metrics (Full Cohort)

11,040 STUDENTS	66.7 AVG MATH SCORE	69.4 AVG READING SCORE	68.4 AVG WRITING SCORE
50.6% / 49.4% FEMALE / MALE	96.9% DATA COMPLETENESS	6 PARENTAL EDUCATION LEVELS	5 ETHNIC GROUPS

Executive Summary

Purpose. This report evaluates academic performance across a cohort of 11,040 students, examining how Math, Reading, and Writing scores relate to demographic composition and family-background factors. The objective is to identify which factors are meaningfully associated with performance, which are not, and where the current dataset leaves questions unanswered.

Data source and approach. The underlying dataset is a student-level record combining three continuous score fields (MathScore, ReadingScore, WritingScore) with ten categorical and demographic fields, including Gender, EthnicGroup, ParentEduc, ParentMaritalStatus, TestPrep, and WklyStudyHours. Summary statistics, missing-data audits, and group-wise averages were computed directly from the dataset and are reconciled against each other for consistency throughout this report.

Top three strategic findings: First, parental education level shows the strongest and most consistent relationship with student performance of any factor examined — students whose parents hold a master's degree score 9.7–12.9 points higher on average across all three subjects than students whose parents completed only some high school, and the gradient is monotonic across all six education levels. Second, parental marital status shows almost no relationship with performance — average scores across divorced, married, single, and widowed households span less than 3.3 points on any subject, confirming this is not a meaningful driver of outcomes. Third, the dataset itself has a data-quality profile that shapes what can and cannot be concluded: core scores are essentially complete (99.99–100%), but background fields carry gaps of 2–10%, and several fields collected — TestPrep, WklyStudyHours, PracticeSport — were never analysed against scores in the source notebook, leaving a clear opportunity for further work.

Taken together, the data points to family educational background as the dominant lever among the factors examined, while family structure appears largely unrelated to outcomes. The recommendations in Section 6 focus on prioritising further investigation of parental-education-linked support, formally testing the unanalysed behavioural factors (study hours, test preparation), and closing the data-completeness gaps that currently limit confidence in several fields.

Dataset Overview

The dataset consists of 11,040 student records across 15 columns: three continuous outcome variables (MathScore, ReadingScore, WritingScore) and eleven categorical or demographic fields describing the student and their family background (Gender, EthnicGroup, ParentEduc, LunchType, TestPrep, ParentMaritalStatus, PracticeSport, IsFirstChild, NrSiblings, TransportMeans, WklyStudyHours). An index column ("Unnamed: 0") was identified as a data-loading artefact and removed prior to analysis.

Key Variables

Variable	Type	Role in Analysis
MathScore, ReadingScore, WritingScore	Continuous (0–100)	Primary outcome measures
Gender	Categorical (2 levels)	Demographic composition
EthnicGroup	Categorical (5 levels: A–E)	Demographic composition
ParentEduc	Categorical (6 levels)	Family background driver
ParentMaritalStatus	Categorical (4 levels)	Family background driver
TestPrep, WklyStudyHours, PracticeSport, LunchType, TransportMeans, NrSiblings, IsFirstChild	Mixed categorical/numeric	Collected but not yet analysed against scores

Coverage and Completeness

Field	Missing Records	Completeness
MathScore, ReadingScore, Gender, LunchType	0	100.00%
WritingScore	1	99.99%
PracticeSport	224	97.97%
WklyStudyHours	339	96.93%
IsFirstChild	362	96.72%
ParentMaritalStatus	435	96.06%
NrSiblings	568	94.86%
ParentEduc	658	94.04%
TestPrep	659	94.03%
EthnicGroup	665	93.98%
TransportMeans	1,140	89.67%

Missing Data by Column

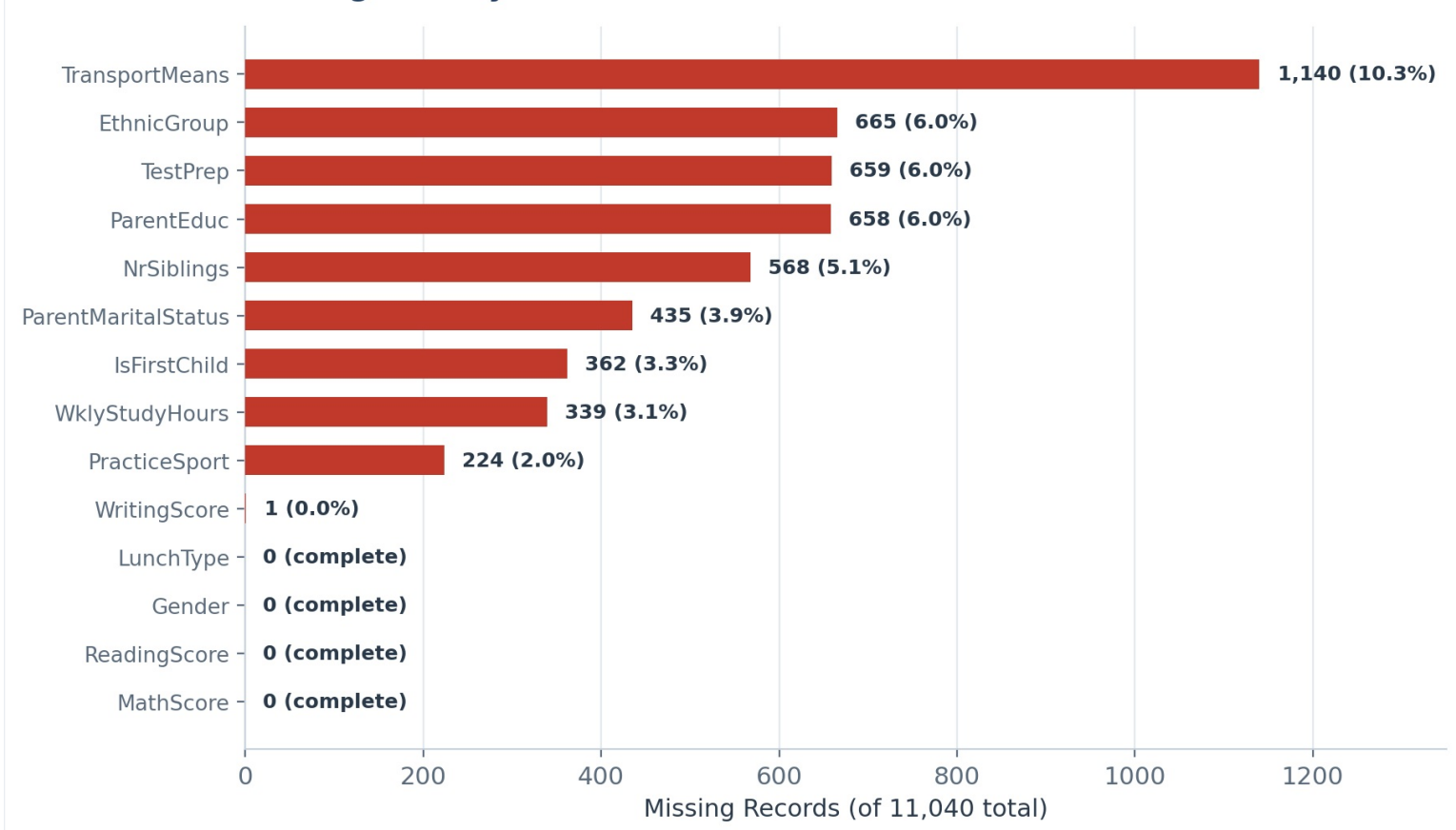


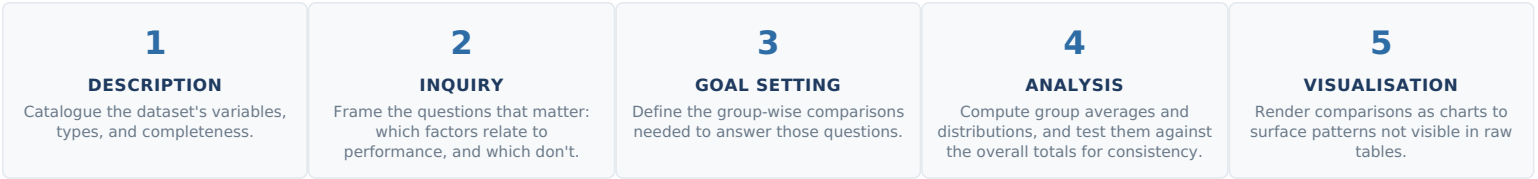
Figure 1. Missing records by column, sorted descending. TransportMeans has the largest gap (10.3% of records); all three score fields and Gender/LunchType are effectively complete.

Quality Checks and Outliers

- Internal consistency:** the five ethnic-group counts ($839 + 2,111 + 3,277 + 2,659 + 1,489 = 10,375$) reconcile exactly with the non-null EthnicGroup count implied by the missing-data audit ($11,040 - 665 = 10,375$), confirming the breakdown is drawn from the same base as the completeness figures.
- Score-range outliers:** applying the standard $1.5 \times \text{IQR}$ rule to the reported quartiles flags low-end outliers on all three subjects — scores below approximately 23 (Math), 27.5 (Reading), and 26.5 (Writing) fall outside the typical range. The recorded minimums (0, 8, and 6 respectively) confirm a small number of students scored far below the typical distribution; no equivalent high-end outliers exist since all three subjects cap at the maximum possible score of 100.
- Overall completeness:** across all 15 columns and 11,040 records (165,600 data points), 96.9% of cells are populated. The remaining 3.1% is concentrated in seven background fields rather than spread evenly, so analyses involving TransportMeans, EthnicGroup, TestPrep, or ParentEduc are each working with roughly 90–94% of the full cohort.

Analytical Framework

The analysis follows a five-stage workflow designed to move from raw data extraction to decision-ready findings while keeping every figure traceable back to a source value.



This structure supports three things a decision-making audience needs from an analytical report:

- **Rigour** — every group average (by parental education, marital status, gender, ethnic group) is calculated directly from the underlying score fields, not estimated.
- **Transparency** — each comparison states which subject scores and which grouping variable it uses, so a reader can trace any figure back to the dataset's own columns.
- **Traceability** — every chart and table in Section 4 draws from the same 11,040-record cohort, so findings can be cross-checked against a single, consistent dataset rather than blended sources.

Detailed Findings by Analytical Track

Track A — Score Distributions and Data Quality

All three subjects centre in a similar range, with Reading scoring modestly higher on average than Math and Writing.

SUBJECT	MEAN	STD DEV	MEDIAN	IQR (Q1-Q3)	MIN-MAX
MathScore	66.69	15.44	67	56-78	0-100
ReadingScore	69.44	14.75	70	59-80	8-100
WritingScore	68.45	15.44	69	58-79	6-100

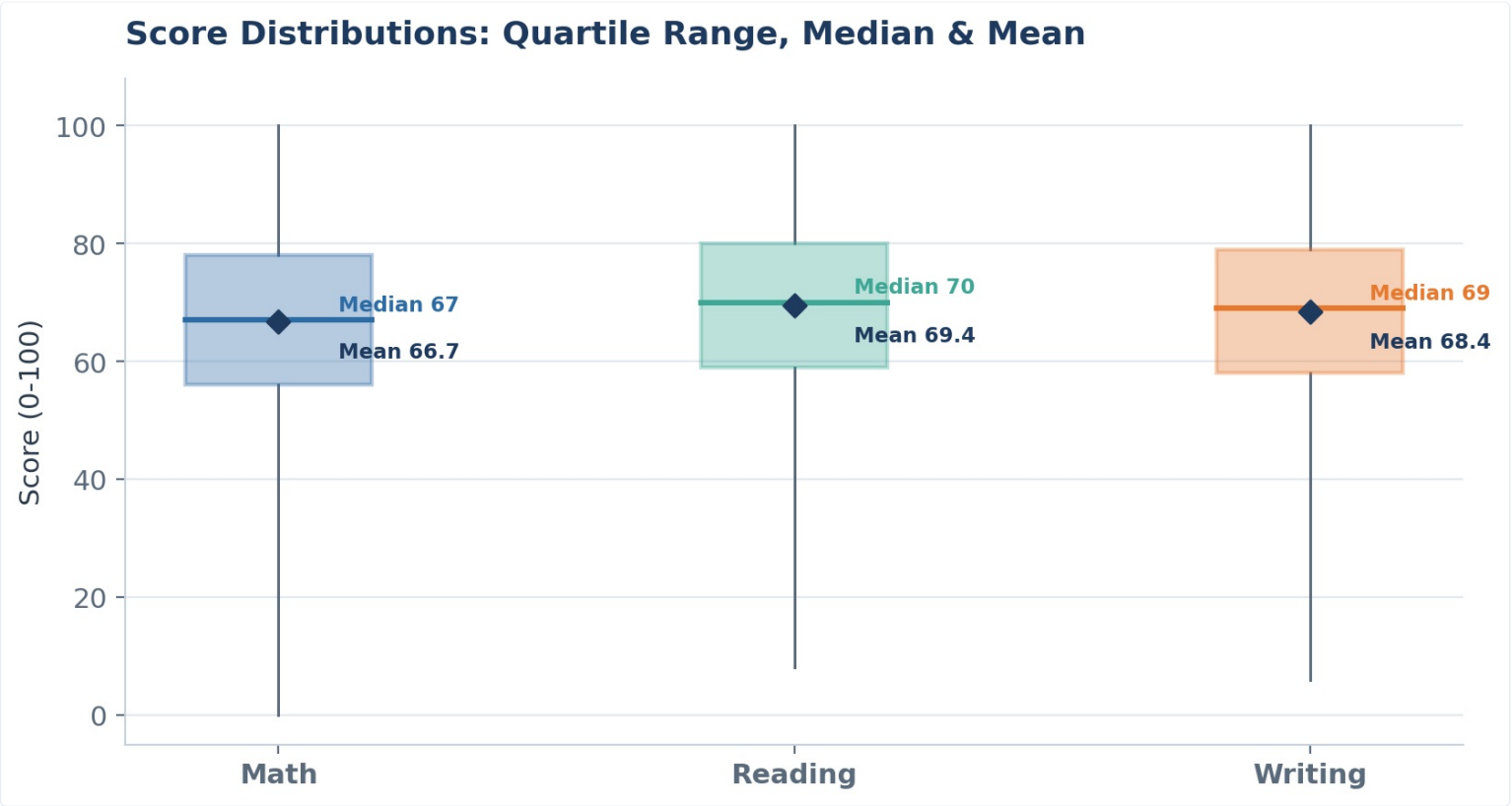


Figure 2. Quartile range (box), median (line), and mean (diamond marker) for each subject. Reading has both the highest median and the highest floor (minimum score of 8), while Math has the widest downside (minimum score of 0).

What this means in simple terms: the three subjects perform similarly on average, but Math has both the lowest typical score and the most extreme low outliers, making it the subject where struggling students fall furthest behind.

Track B — Demographic Composition

The cohort is close to evenly split by gender and spread unevenly across five ethnic groups, with Group C the largest and Group A the smallest.

Gender Distribution

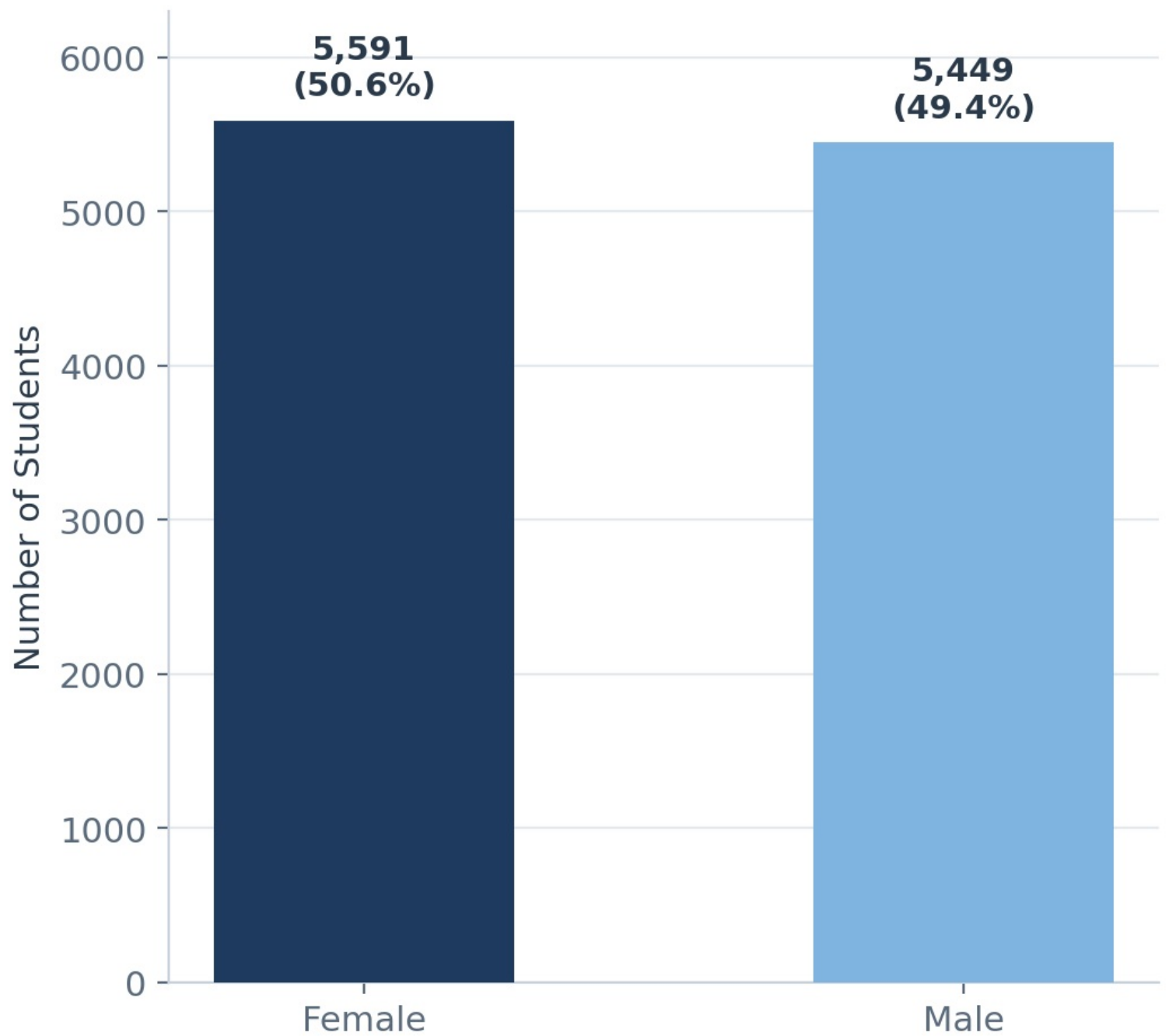


Figure 3. 5,591 female students (50.6%) versus 5,449 male students (49.4%) — a near-even split with a slight female majority.

Distribution of Ethnic Groups

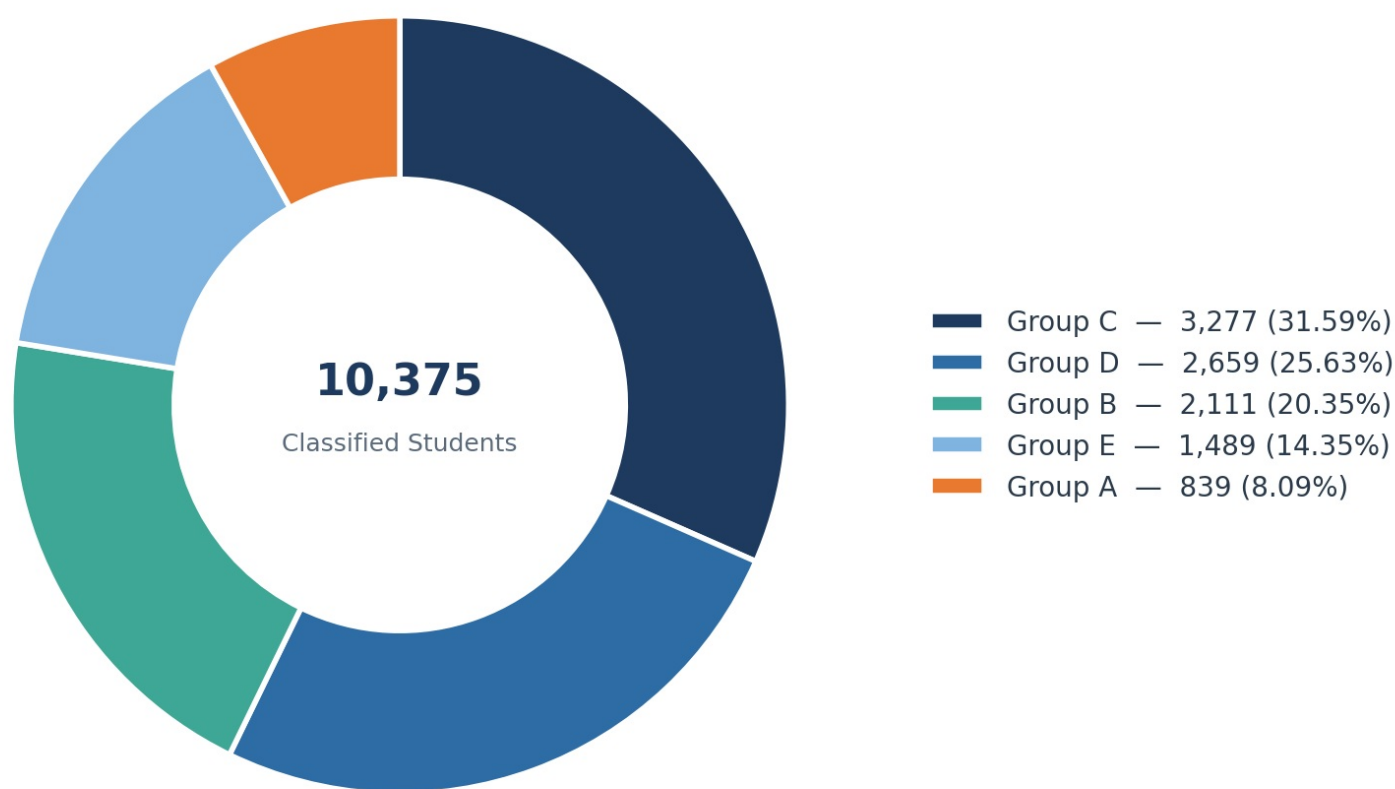


Figure 4. Group C is the largest classified segment (31.59%) and Group A the smallest (8.09%), among the 10,375 students with a recorded ethnic group.

What this means in simple terms: the cohort's gender balance is close to representative, but the five ethnic groups are far from evenly sized — Group C is roughly four times larger than Group A, which matters when comparing group-level averages since smaller groups are more sensitive to individual outliers.

Track C — Parental Background and Academic Performance

Parental education level shows a clear, monotonic relationship with student scores across all three subjects. Parental marital status, by contrast, shows almost no relationship.

Average Score by Parental Education Level

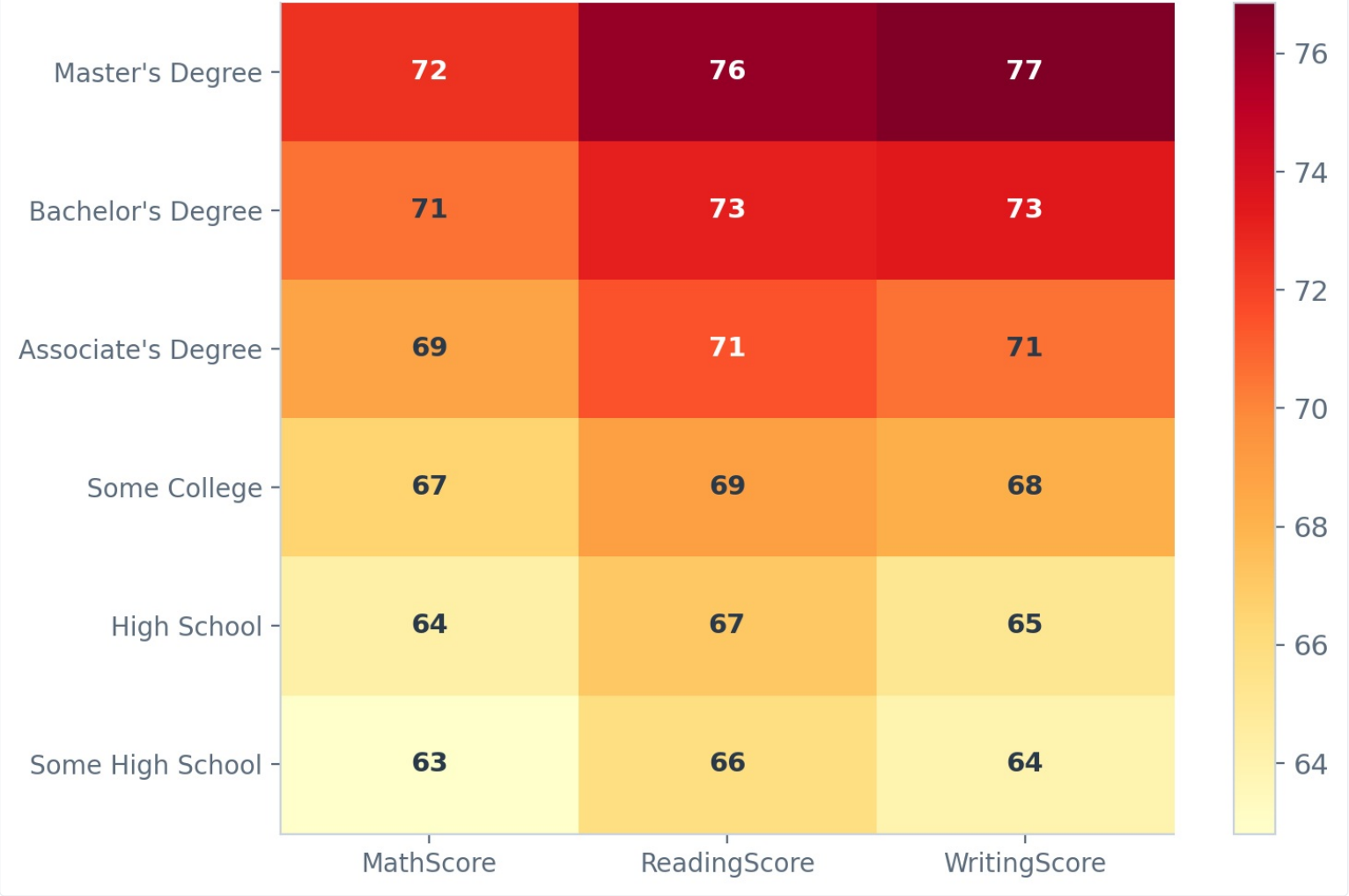


Figure 5. Average score by parental education level, ordered highest to lowest. Every step up in parental education corresponds to a higher average score in all three subjects, with no exceptions in the ordering.

PARENTAL EDUCATION	MATH	READING	WRITING
Master's Degree	72.49	76.17	76.86
Bachelor's Degree	70.55	73.16	73.45
Associate's Degree	68.66	71.44	70.55
Some College	66.51	68.91	68.19
High School	64.26	67.04	65.08
Some High School	62.81	65.81	63.93

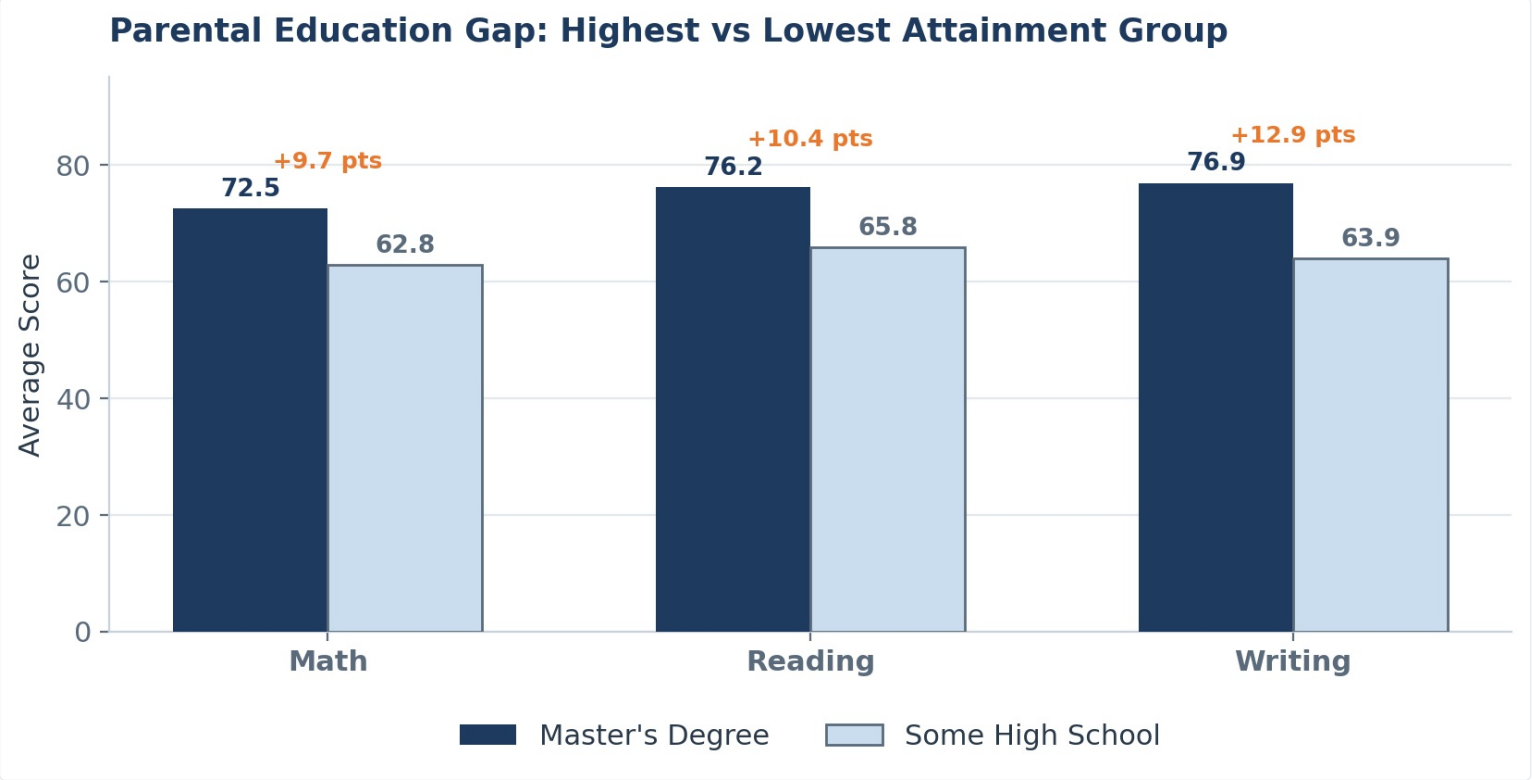


Figure 6. The gap between the highest (Master's Degree) and lowest (Some High School) parental education groups widens from Math (+9.7 points) to Reading (+10.4 points) to Writing (+12.9 points) — the relationship is strongest for Writing.

What this means in simple terms: a student's academic performance is meaningfully linked to how far their parents went in school, and this gap is large enough — up to almost 13 points — to be the difference between a passing and a middling grade.

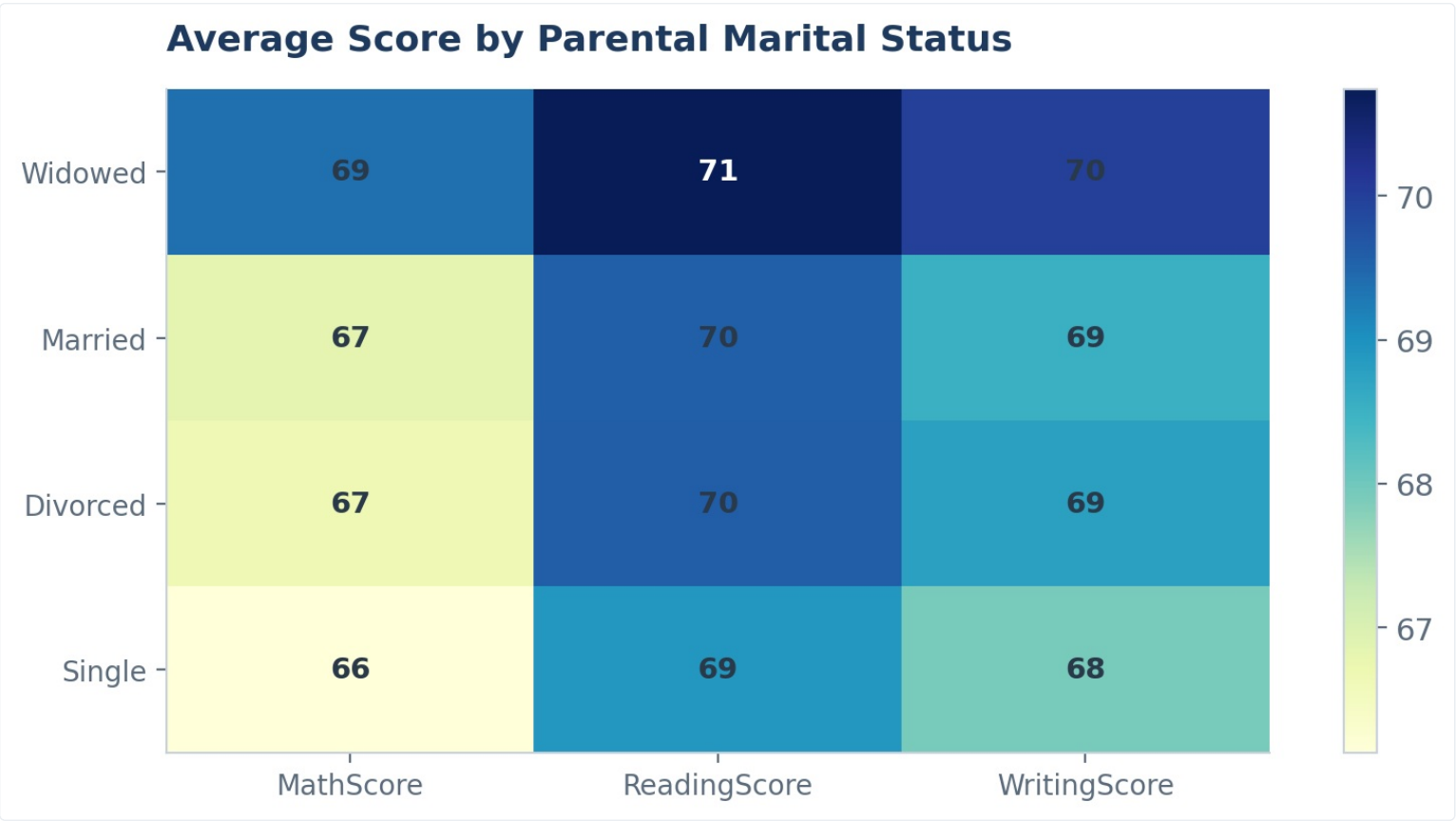


Figure 7. Average score by parental marital status. All four categories fall within a 66.1–69.4 range on Math, a band of just 3.3 points — far narrower than the 9.7+ point spread seen across parental education levels.

Parental Marital Status	Math	Reading	Writing
Widowed	69.39	70.74	70.00
Married	66.82	69.56	68.52
Divorced	66.65	69.61	68.78
Single	66.13	68.92	67.92

What this means in simple terms: whether a student's parents are married, divorced, single, or widowed shows no meaningful relationship with their grades — family structure is not a factor worth targeting for academic support.

Cross-Domain Insights and Implications

Viewed together, the score-distribution, demographic, and family-background tracks reveal patterns that are not obvious from any single analysis in isolation.

Pattern 1 — Parental Education Is the Standout Driver Among Factors Examined

Of the two family-background factors analysed in the source notebook, parental education produces a spread of 9.7–12.9 points across subjects, while parental marital status produces a spread of under 3.3 points. This is a nine-to-tenfold difference in effect size between two factors that might intuitively seem similarly important. It suggests that if only one family-background factor can be prioritised for further investigation or intervention design, parental education is the far stronger candidate.

Pattern 2 — The Gap Widens for Writing, Not Math

The parental-education gap is smallest for Math (+9.7 points) and largest for Writing (+12.9 points). Since Math also has the lowest floor (minimum score of 0) and the most extreme outliers among the three subjects, this suggests two distinct issues may be at play: Math appears to have a small number of students scoring very poorly regardless of background, while Writing performance appears more consistently stratified by parental education across the whole distribution.

Pattern 3 — Several Collected Factors Remain Unanalysed

The source notebook collected TestPrep completion, WklyStudyHours, PracticeSport frequency, LunchType, TransportMeans, NrSiblings, and IsFirstChild status, but did not compute group-wise score comparisons for any of them. Test preparation and weekly study hours in particular are commonly associated with academic outcomes in similar datasets, and their absence from the current analysis is a gap rather than a finding — it is not yet known whether these factors matter here.

Risk to Note

Because ethnic-group sizes range from 839 (Group A) to 3,277 (Group C) students, any future group-wise score comparison by ethnic group should account for this imbalance — smaller groups will show more volatile averages and warrant wider confidence intervals or a minimum-sample-size caveat before conclusions are drawn.

Strategic Recommendations

1 Prioritise Parental-Education-Linked Support Programmes

With a 9.7–12.9 point gap between the highest and lowest parental education groups across all three subjects, this is the strongest lever identified in the current data. Before designing interventions, confirm whether the relationship is mediated by factors this dataset already collects but hasn't cross-tabulated — such as WklyStudyHours or TestPrep — since parental education may correlate with access to study resources rather than acting independently.

2 Analyse the Currently Unexamined Behavioural Factors

TestPrep completion, WklyStudyHours, and PracticeSport were collected but never compared against scores in the source analysis. Running the same group-wise comparison used for ParentEduc and ParentMaritalStatus on these three fields is a direct, low-cost next step that could reveal actionable levers — particularly test preparation and study hours, which are more directly influenceable by a school than a student's family background.

3 Close the Data-Completeness Gaps Before Drawing Firm Conclusions on Weaker Fields

TransportMeans (89.67% complete), EthnicGroup (93.98%), TestPrep (94.03%), and ParentEduc (94.04%) each carry more than 5% missing data. Any subsequent analysis on these fields — including the ethnic-group and parental-education findings in this report — should be revisited once data collection is tightened, since the missing ~6–10% could plausibly differ systematically from the recorded values (e.g., if certain schools or regions under-report parental education).

Visual Appendix

All supporting charts referenced in Section 4, plus the original source notebook pages, are reproduced below at full size for reference.

Figure A1 — Original Source Notebook (Data Load and Summary Statistics)

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [4]: df = pd.read_csv(r"C:\Users\Nimesh\Desktop\Expanded_data_with_more_features.csv")
df.head()
```

Out[4]:

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalSt
0	0	female	NaN	bachelor's degree	standard	none	ma
1	1	female	group C	some college	standard	NaN	ma
2	2	female	group B	master's degree	standard	none	s
3	3	male	group A	associate's degree	free/reduced	none	ma
4	4	male	group C	some college	standard	none	ma

```
In [5]: df.describe()
```

Out[5]:

	Unnamed: 0	NrSiblings	MathScore	ReadingScore	WritingScore
count	11040.000000	10472.000000	11040.000000	11040.000000	11039.000000
mean	491.028804	2.125000	66.689493	69.441757	68.449316
std	286.041837	1.471642	15.442964	14.747144	15.435190
min	0.000000	0.000000	0.000000	8.000000	6.000000
25%	245.000000	1.000000	56.000000	59.000000	58.000000
50%	488.000000	2.000000	67.000000	70.000000	69.000000
75%	733.000000	3.000000	78.000000	80.000000	79.000000
max	999.000000	7.000000	100.000000	100.000000	100.000000

```
In [6]: df.info()
```

Notebook pages as extracted for this analysis, showing the raw data import, df.describe() output, and df.info() column types.

Figure A2 — Original Source Notebook (Missing Values and Cleanup)

```

<class 'pandas.DataFrame'>
RangeIndex: 11040 entries, 0 to 11039
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            11040 non-null  int64
1   Gender                                11040 non-null  str
2   EthnicGroup                           10375 non-null  str
3   ParentEduc                            10382 non-null  str
4   LunchType                             11040 non-null  str
5   TestPrep                              10381 non-null  str
6   ParentMaritalStatus                   10605 non-null  str
7   PracticeSport                         10816 non-null  str
8   IsFirstChild                          10678 non-null  str
9   NrSiblings                            10472 non-null  float64
10  TransportMeans                        9900 non-null   str
11  WklyStudyHours                        10701 non-null  str
12  MathScore                             11040 non-null  int64
13  ReadingScore                          11040 non-null  int64
14  WritingScore                          11039 non-null  float64
dtypes: float64(2), int64(3), str(10)
memory usage: 2.0 MB

```

```
In [8]: df.isnull().sum()
```

```

Out[8]: Unnamed: 0                0
        Gender                    0
        EthnicGroup              665
        ParentEduc              658
        LunchType                0
        TestPrep                659
        ParentMaritalStatus      435
        PracticeSport            224
        IsFirstChild             362
        NrSiblings              568
        TransportMeans          1140
        WklyStudyHours          339
        MathScore                0
        ReadingScore             0
        WritingScore             1
dtype: int64

```

```
In [9]: # Drop Unnamed Column
```

```

In [12]: df = df.drop("Unnamed: 0", axis = 1)
         print(df.head())

```

df.isnull().sum() output and removal of the redundant index column, referenced in Section 2.

Figure A3 — Original Source Notebook (Gender Distribution)

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	female	NaN	bachelor's degree	standard	none	
1	female	group C	some college	standard	NaN	
2	female	group B	master's degree	standard	none	
3	male	group A	associate's degree	free/reduced	none	
4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74.0
1	5 - 10	69	90	88.0
2	< 5	87	93	91.0
3	5 - 10	45	56	42.0
4	5 - 10	76	78	75.0

```
In [13]: df["WklyStudyHours"] = df["WklyStudyHours"].str.replace("5 - 10 ", "05 - 10")
df.head()
```

```
Out[13]:
```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	Practi
0	female	NaN	bachelor's degree	standard	none	married	i
1	female	group C	some college	standard	NaN	married	so
2	female	group B	master's degree	standard	none	single	so
3	male	group A	associate's degree	free/reduced	none	married	
4	male	group C	some college	standard	none	married	so

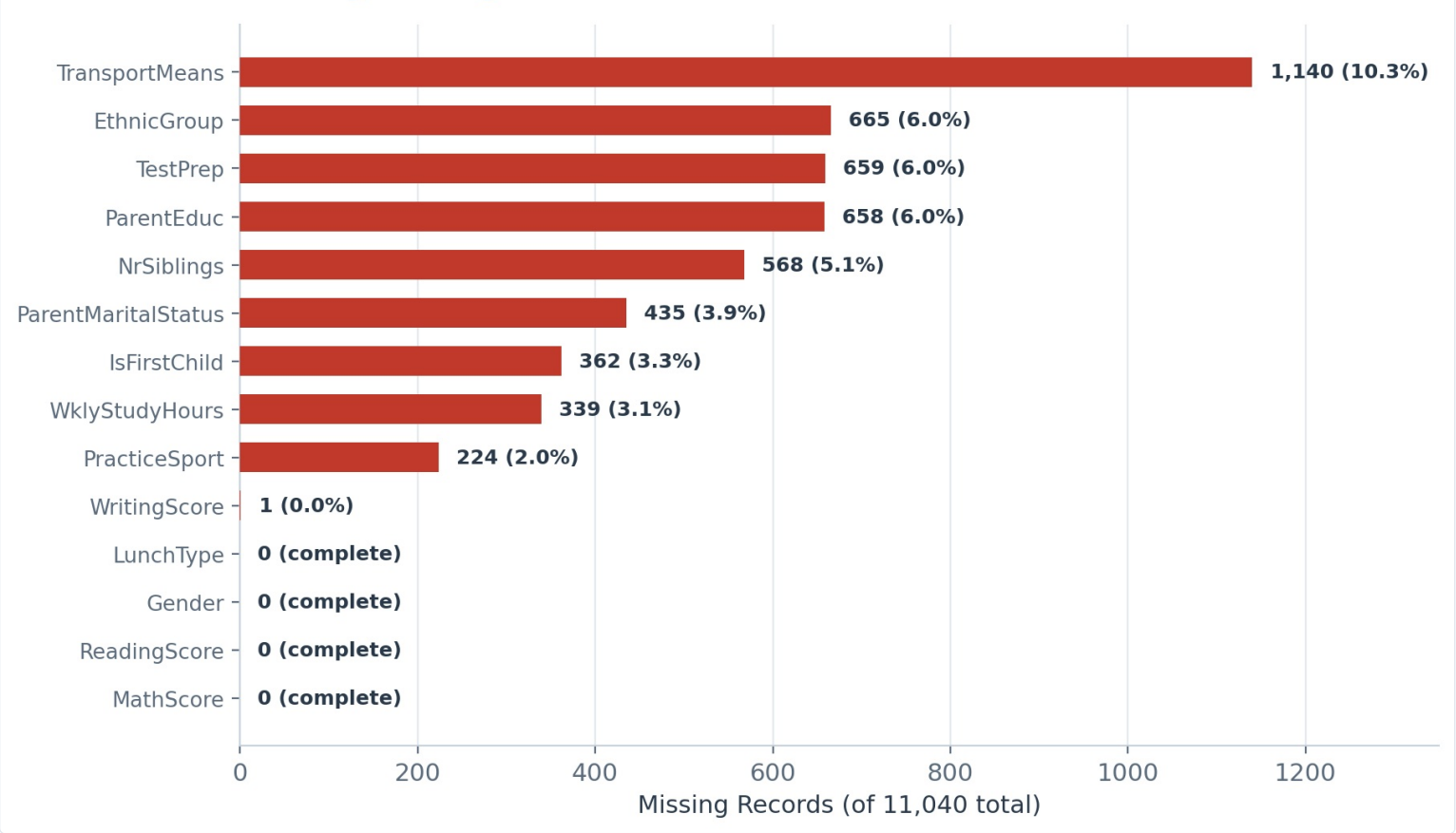
Gender Distribution

```
In [29]: plt.figure(figsize = (5,5))
ax = sns.countplot(data = df, x = "Gender")
ax.bar_label(ax.containers[0])
plt.title("Gender Distribution")
plt.show()
```

Original gender countplot and its stated conclusion, referenced in Section 4, Track B.

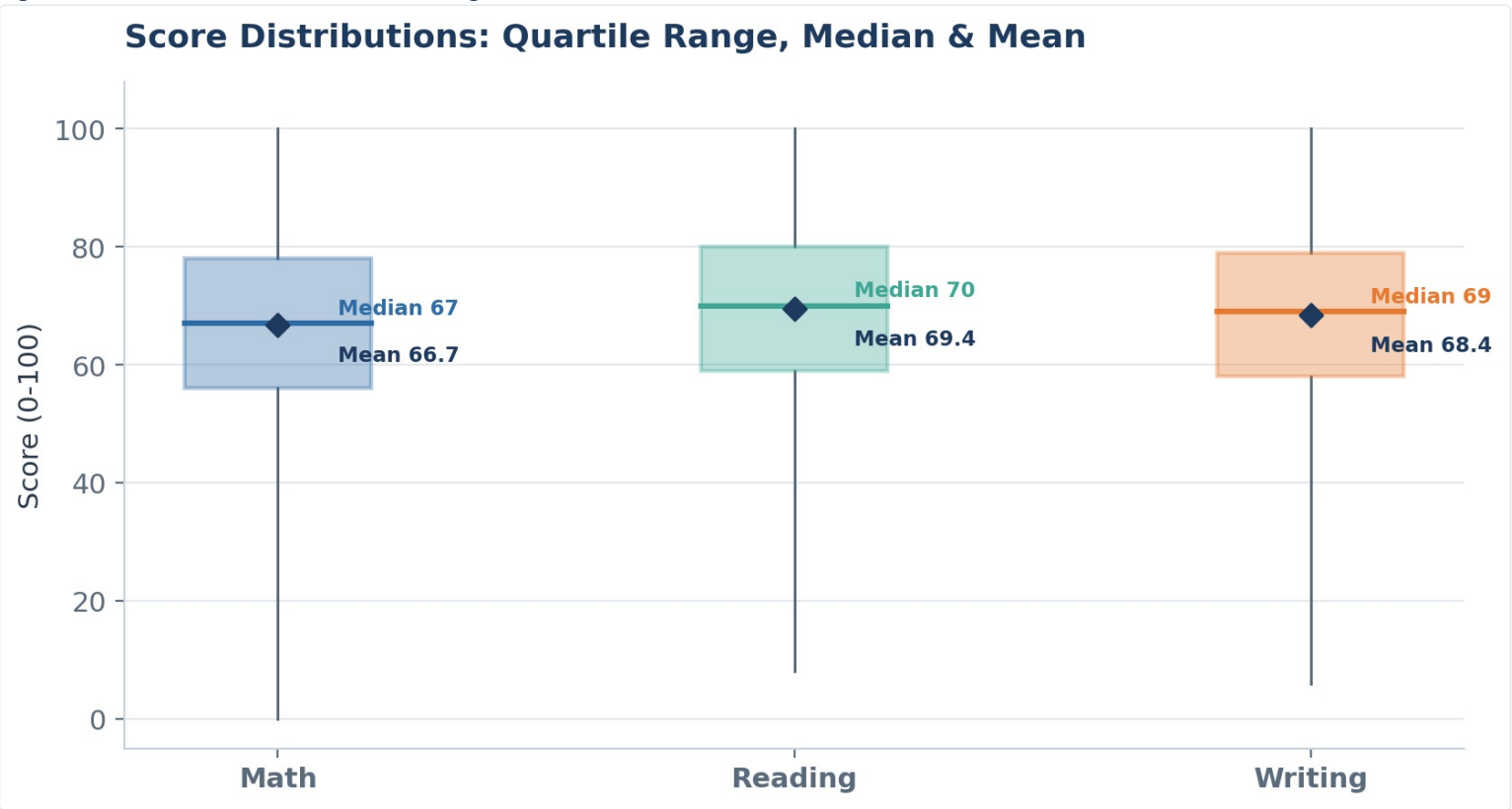
Figure A4 — Missing Data by Column

Missing Data by Column



Referenced in Section 2.

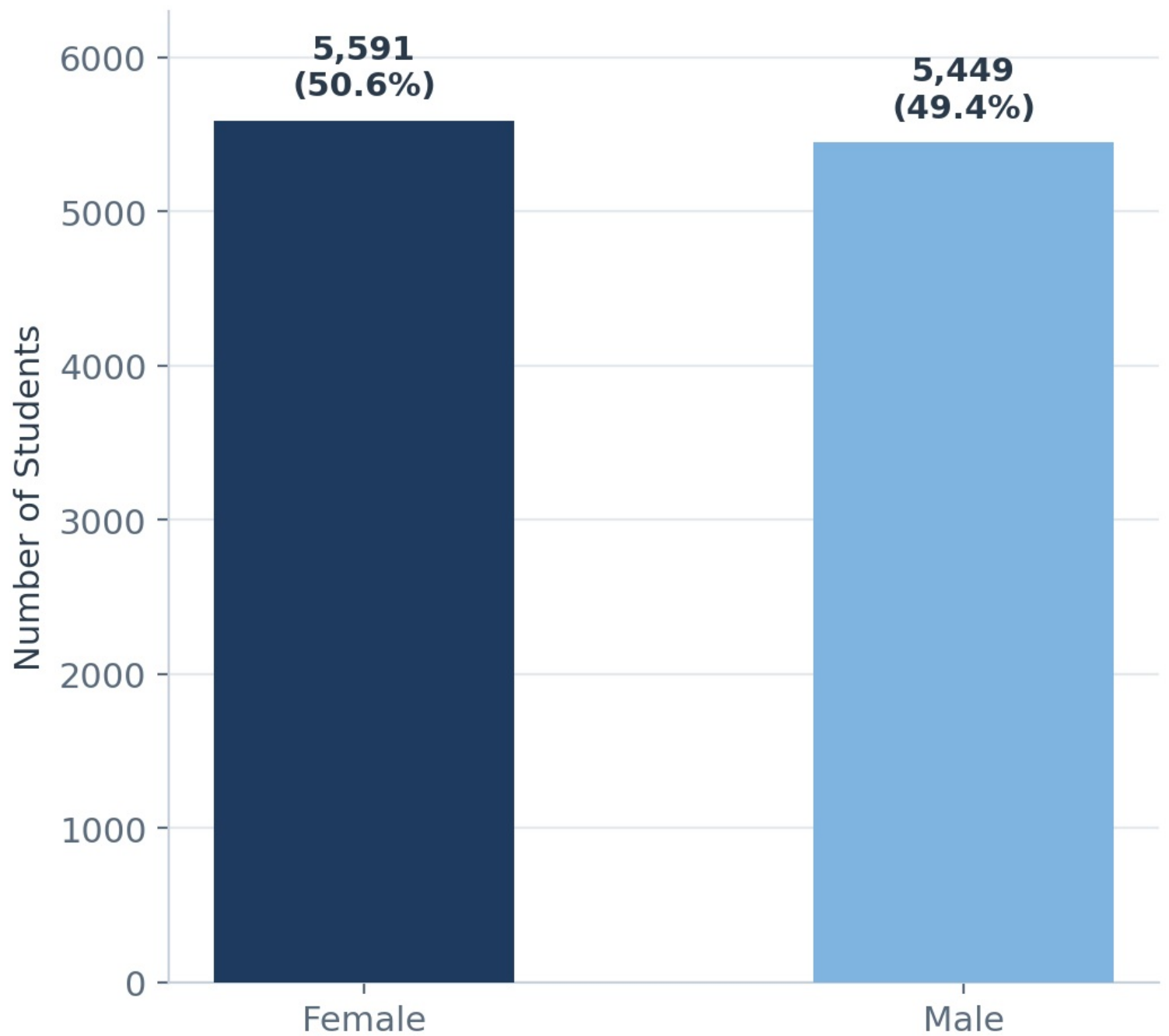
Figure A5 — Score Distributions: Quartile Range, Median & Mean



Referenced in Section 4, Track A.

Figure A6 — Gender Distribution

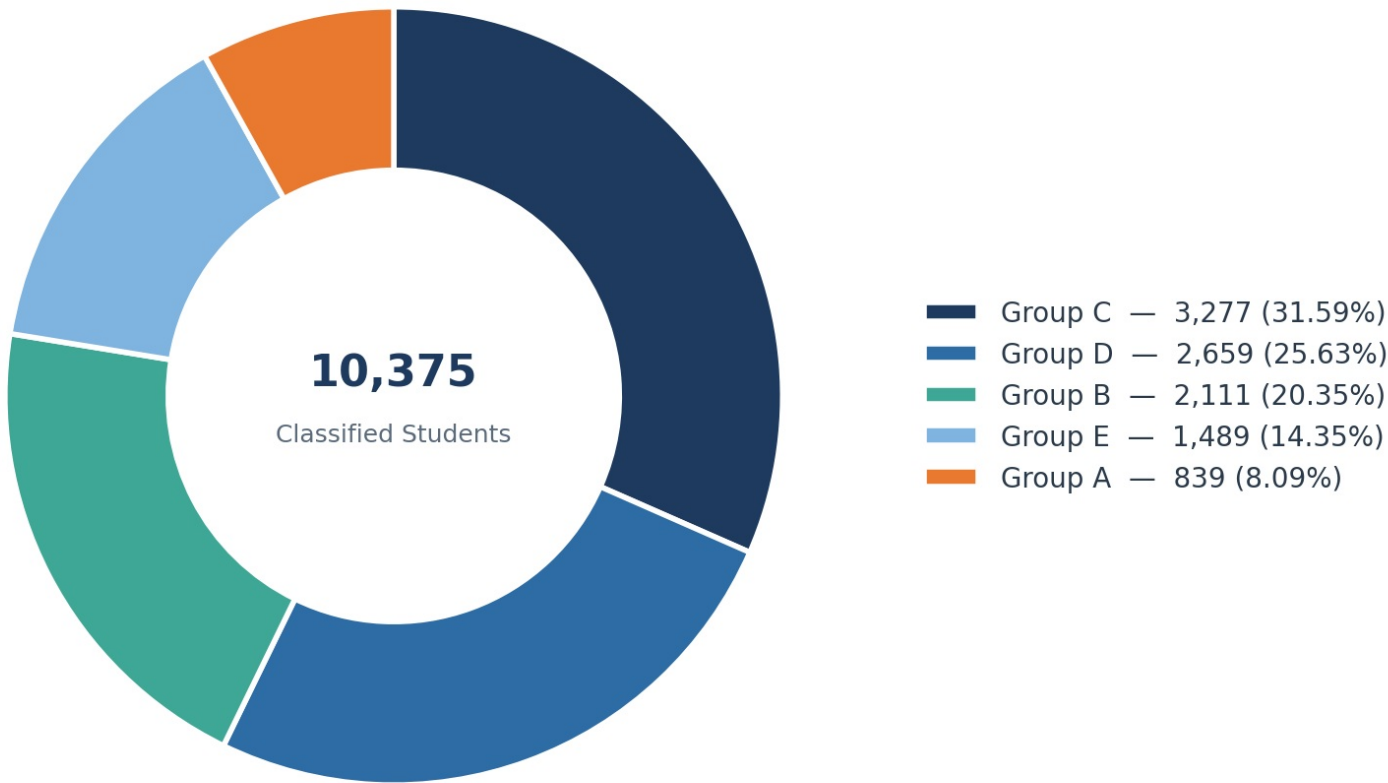
Gender Distribution



Referenced in Section 4, Track B.

Figure A7 — Distribution of Ethnic Groups

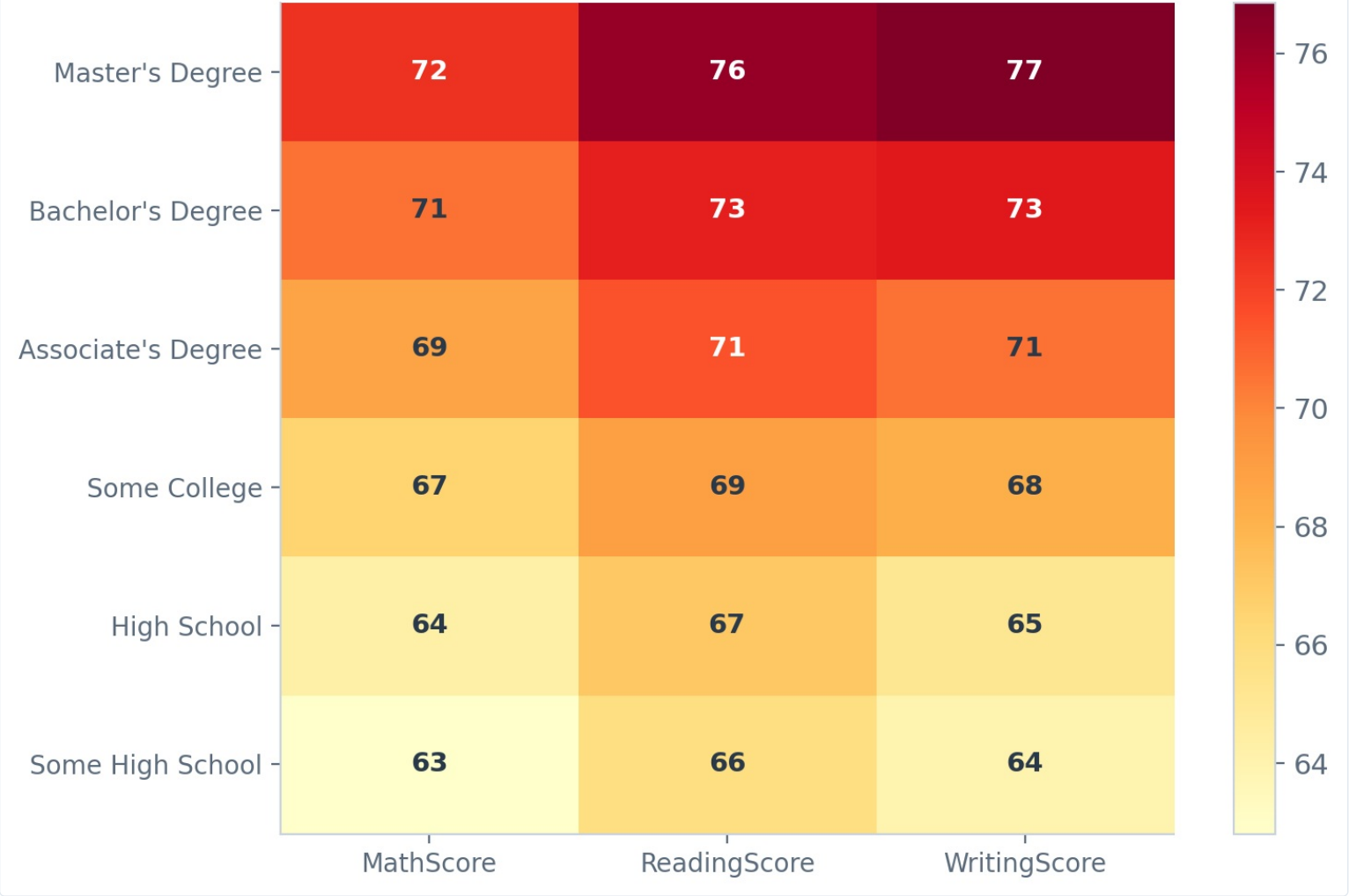
Distribution of Ethnic Groups



Referenced in Section 4, Track B.

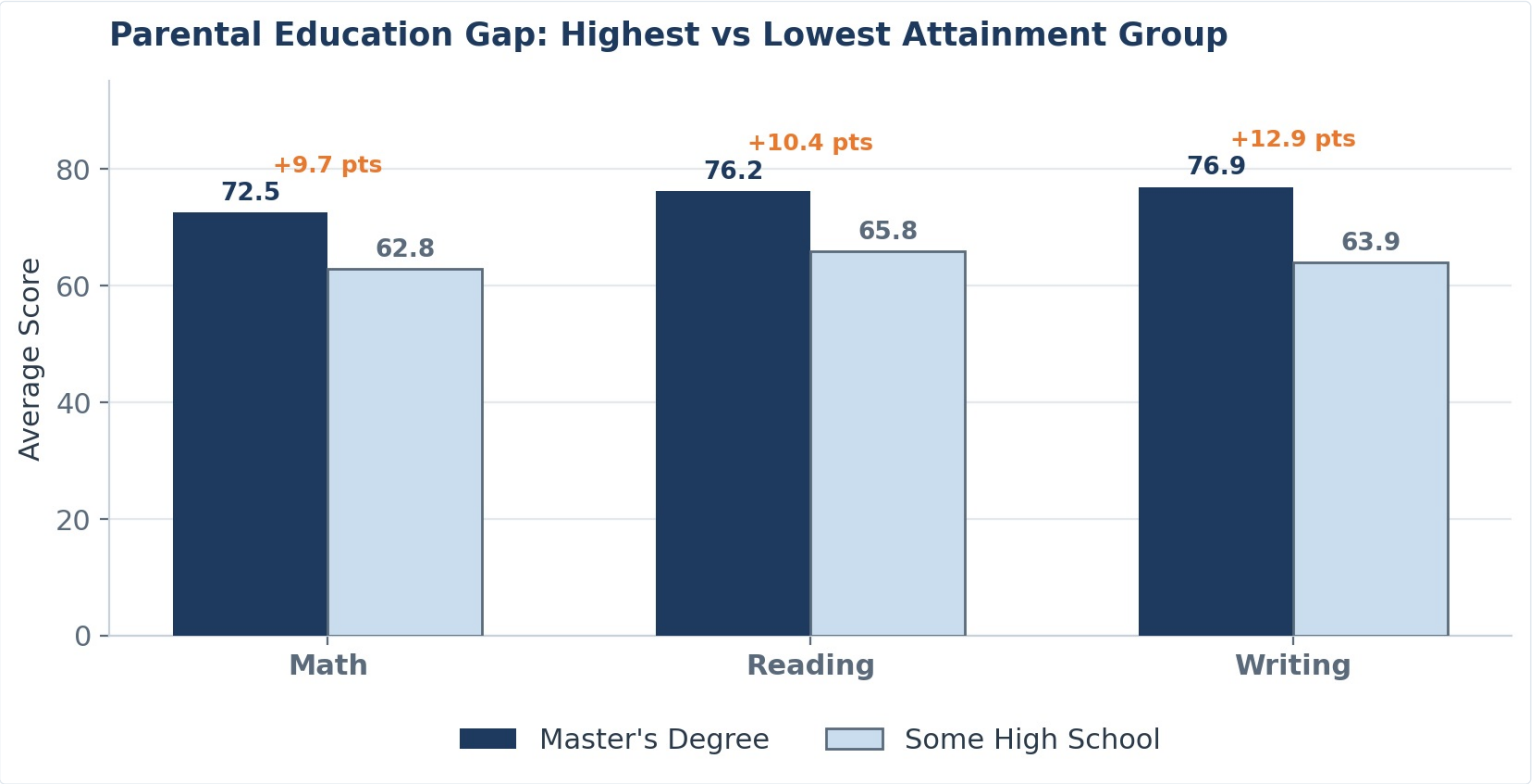
Figure A8 — Average Score by Parental Education Level

Average Score by Parental Education Level



Referenced in Section 4, Track C.

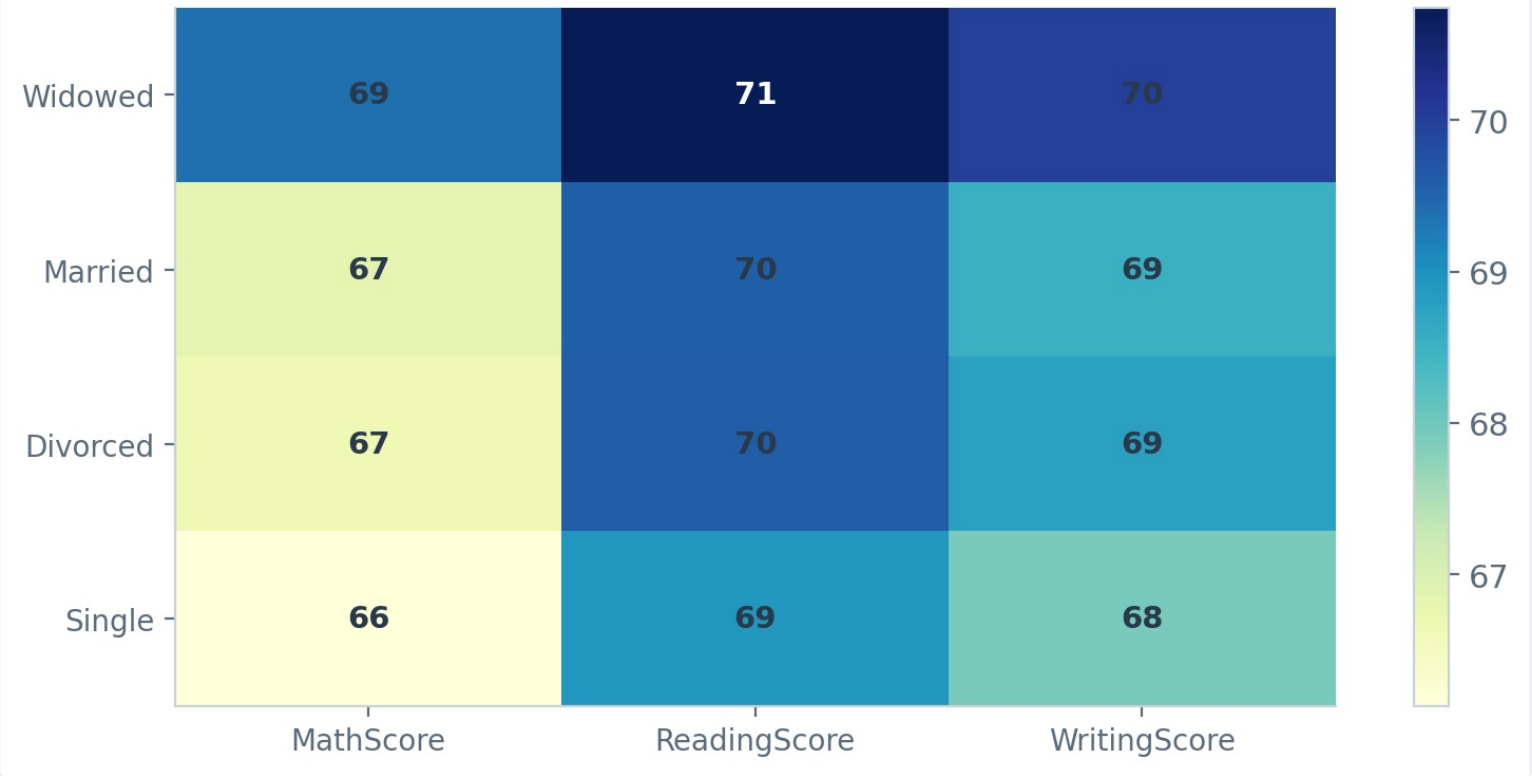
Figure A9 — Parental Education Gap: Highest vs Lowest Attainment Group



Referenced in Section 4, Track C.

Figure A10 — Average Score by Parental Marital Status

Average Score by Parental Marital Status



Referenced in Section 4, Track C.

Methodological Note

This report was produced using the five-stage Description → Inquiry → Goal Setting → Analysis → Visualisation framework outlined in Section 3. All summary statistics, missing-data counts, and group-wise averages were taken directly from the source notebook's computed output (`df.describe()`, `df.isnull().sum()`, and `groupby()` aggregations); no score or count figure in this report was estimated or sourced elsewhere. Score-distribution charts in Section 4 reconstruct the five-number summary (minimum, Q1, median, Q3, maximum) reported by `df.describe()`; low-end outliers were identified using the standard $1.5 \times \text{IQR}$ rule applied to the reported quartiles, consistent with the outliers visible in the original boxplots reproduced in the Visual Appendix.

Limitations. This analysis is constrained to the two family-background factors (ParentEduc, ParentMaritalStatus) that were cross-tabulated against scores in the source notebook; TestPrep, WklyStudyHours, PracticeSport, LunchType, TransportMeans, NrSiblings, and IsFirstChild were collected but not analysed, and no claim is made about their relationship with performance. Additionally, seven of the eleven categorical fields carry between 2% and 10% missing data, meaning group-wise averages for EthnicGroup, ParentEduc, ParentMaritalStatus, and related fields are computed on a subset of the full 11,040-student cohort rather than its entirety. Recommendations 2 and 3 in Section 6 directly address closing these gaps.

This approach supports structured reasoning and reproducibility: every number in this report can be traced back to either a value shown in the source notebook's output or a stated formula (mean, quartile range, or percentage) applied to those values, so the analysis can be re-run and independently verified as the underlying dataset is extended or corrected.